

1.5 Troubleshooting Hardware Device Conflicts In Windows

Hardware devices can conflict with each other due to a variety of reasons. Every hardware device installed must be allocated a set of operating system (Windows) resources to operate correctly. These resources include IRQ, DMA (Direct Memory Access), I/O port addresses and memory resources. Some of these resources will be shared by more than one device while others cannot. This depends on the capabilities of the hardware and its software drivers.

Each device installed will have one or more software device drivers that enable the operating system to access and use the device. The software could be provided either by Microsoft itself that will enable basic functionality, or by the device vendor who knows more about the specific model and hence will be able to provide a more tightly integrated device driver.

Normally, Windows Plug and Play will automatically detect the newly-added hardware, determine the system resources required and assign them properly. Plug and Play will also search for the required software drivers and install them transparently. If Plug and Play cannot find the required drivers, it will prompt the user to provide them by inserting the CD/floppy provided by the device manufacturer. Additionally, it will also reconfigure the system resources if required, for example when the newly added device requires some resources already in use.

1.5.1 Device Conflict Symptoms

A hardware conflict may occur when you add a new device to your system. In some instances, this conflict will not cause any problem, and depending on the operating system you may not even be aware that such a conflict exists. Typical symptoms of hardware conflicts include:

- The operating system may freeze frequently
- The new device may suffer performance problems